

Mancala Game Use Cases

Use Case 1: Start Game

Step	User's Action	System's Response
1	The user runs the Mancala program.	
2		The system displays the initial setup screen with board style choices.
3	The user selects one of the board styles.	
4		The system saves the selected board style and asks for the number of stones per pit.
5	The user chooses either 3 or 4 stones per pit.	
6		The system creates the Mancala board with the selected style and number of stones.
7		The system displays the board, pits, Mancalas, stones, and labels.
8		The system sets Player A as the first player and displays Player A's turn.

Use Case 2: Select a Pit

Step	User's Action	System's Response
1	The current player clicks one of their pits.	
2		The system receives the selected pit index.
3		The system checks whether the selected pit is valid for the current player.
4		If the move is valid, the system removes all stones from the selected pit.
5		The system distributes the stones counter-clockwise around the board.
6		The system skips the opponent's Mancala if needed.
7		The system checks whether the last stone causes a capture or extra turn.
8		The system updates the board display and stone counts.
9		The system updates whose turn it is if the player does not get an extra turn.

Use Case 3: Undo Move

Step	User's Action	System's Response
1	The current player clicks the Undo button.	
2		The system checks whether undo is allowed.

3		If undo is allowed, the system restores the board state from before the last move.
4		The system restores the previous player's turn.
5		The system updates the board display and stone counts.
6	The player clicks Undo again without making another move.	
7		The system does not allow another undo and leaves the board unchanged.

Use Case 4: Invalid Pit Selection

Step	User's Action	System's Response
1	The current player clicks an invalid pit, such as an empty pit, opponent's pit, or Mancala.	
2		The system receives the selected pit index.
3		The system checks whether the selected pit is valid.
4		The system rejects the move because the selected pit is not allowed.
5		The board state does not change.
6		The system continues displaying the same player's turn.

Use Case 5: End Game

Step	User's Action	System's Response
1	The current player makes a move.	
2		The system distributes the stones and updates the board.
3		The system checks whether all six pits on one player's side are empty.
4		If one side is empty, the system ends the game.
5		The system moves all remaining stones from the other side into that player's Mancala.
6		The system compares the stones in both Mancalas.
7	The players view the result.	
8		The system displays the winner or shows that the game is a tie.